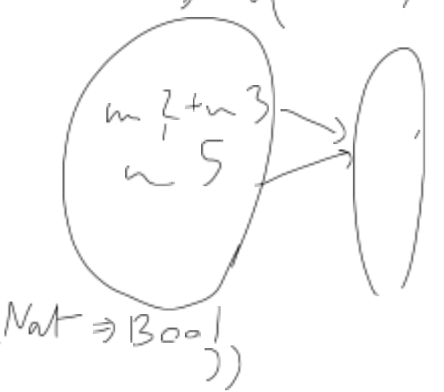
$$\begin{tikzcd} AST \arrow[r, "infer"] & WT \arrow[l, bend left, "erase"] \end{tikzcd}$$

$WT_{+=}$) infer : U.Tm → opt{ A:Ty & Tm A }

(infer (nu 2 + nu 3) = Some (Nat, nu 5)

$$\boxed{\begin{tikzcd} AST \arrow[r, "infer"] & WT_{+=} \arrow[l, bend left, "erase"] \end{tikzcd}}$$ ②

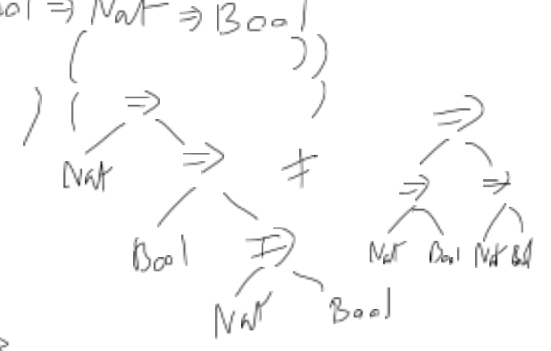


$$\frac{A \quad B}{A \Rightarrow B}$$

lam (Tm x ⇒ t)

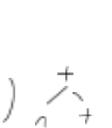
STLC
Ty : Set
Tm : Ty → Set
Bool, Nat, ...
$\_\Rightarrow\_ : Ty \to Ty \to Ty$
$\_\cdot\_ : Tm\,(A \Rightarrow B) \to Tm\,A \to Tm\,B$
lam : (Tm A → Tm B) → Tm (A ⇒ B) ← kötő operátor
β : lam t · a = t a
η : t = lam λx. t · x

Nat → Bool ⇒ Nat ⇒ Bool



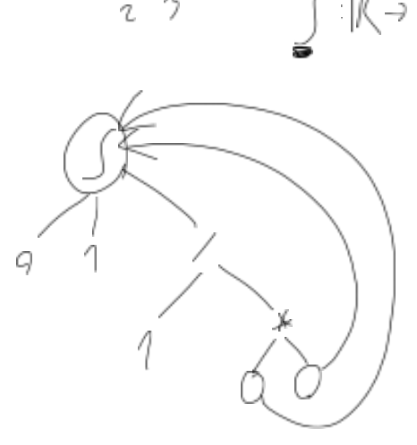
$$\underbrace{\int_0^1 \frac{1}{x^2}}_{\text{kötés hatásköre}} d\overset{\text{kötött változó}}{ⓧ} = \int_0^1 \frac{1}{ⓨ^2} dy$$

1+(2+3)



$\int : \mathbb{R} \to \mathbb{R} \to (\mathbb{R} \to \mathbb{R}) \to \mathbb{R}$

$$\frac{n : \mathbb{R} \quad m : \mathbb{R} \qquad x : \mathbb{R} \vdash t : \mathbb{R}}{\int_n^m t \, dx : \mathbb{R}}$$



$(\lambda x.\ x + x) + y$

$$\frac{A \quad B \qquad x : A \vdash t : B \qquad a : A}{(\mathrm{lam}\ x.t) \cdot a = t[x \mapsto a]}$$

A : Set
con : (A → ⊥) → A

$f : A \to \bot$
$f\,(\mathrm{con}\ g) = g\,(\mathrm{con}\ g)$
$\mathrm{con}\ g : A$, $f : A \to \bot$
lehetetlen : ⊥
lehetetlen = f (con f)

$$\sum_{x=0}^{10} (x+3)$$

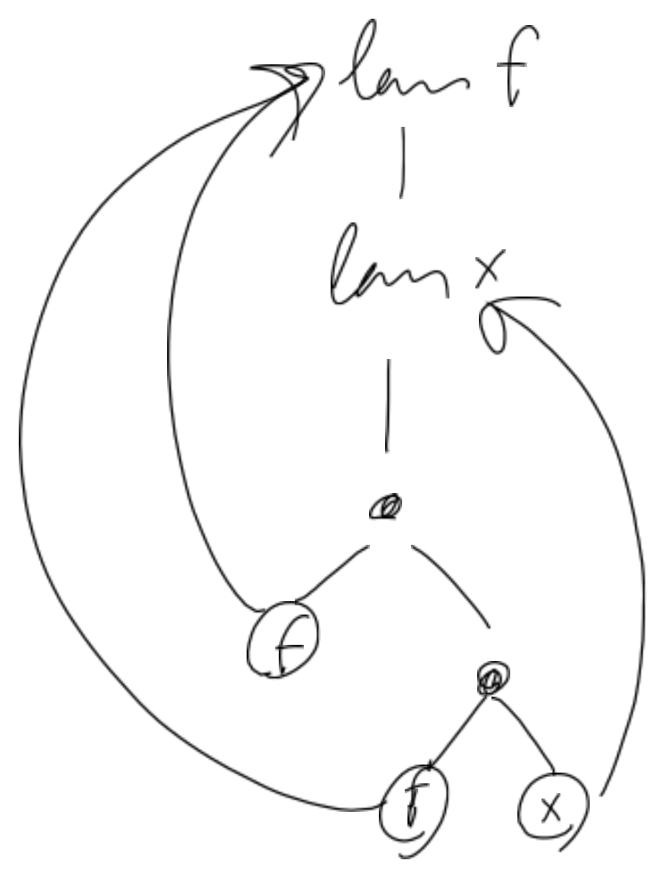
```
for ⓘ := 1 to 10
  x := ⓘ + x
```

```
int fun(int ⓧ) {
  return ⓧ + ⓧ;
}
```

$$\frac{A \quad B \qquad t : A \Rightarrow B}{t = \mathrm{lam}\ x.\ t \cdot x}$$

$$\boxed{\frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \mathrm{la}\ x.t : A \Rightarrow B}}$$

$$\frac{x : A \vdash t : B}{\mathrm{lam}\ x.t : A \Rightarrow B} \qquad \frac{t : A \Rightarrow B \quad u : A}{t \cdot a : B}$$



$$\cfrac{\cfrac{}{f : C \Rightarrow E, x : C \vdash f : E \Rightarrow D} \qquad \cfrac{\cfrac{}{f : C \Rightarrow E, x : C \vdash f : C \Rightarrow E} \qquad \cfrac{}{f : C \Rightarrow E, x : C \vdash x : C}}{f : C \Rightarrow E, x : C \vdash f \cdot x : E}}{\cfrac{f : C \Rightarrow E, x : C \vdash f \cdot (f \cdot x) : D}{\cfrac{f : C \Rightarrow E \vdash \mathrm{lam}\ \overset{x}{\underset{x}{x}}.\ \overset{t}{\underset{t}{f \cdot (f \cdot x)}} : \overset{B}{\underset{A \Rightarrow B}{C \Rightarrow D}}}{\mathrm{lam}\ \underset{x}{f}.\ \underbrace{\mathrm{lam}\ x.\ f \cdot (f \cdot x)}_{t} : (C \Rightarrow C) \Rightarrow C \Rightarrow C}}}$$

$\boxed{C = E = D}$ (with $\overset{x}{}$ $\overset{A}{}$ over $f : C \Rightarrow E$)

$$A = A \Rightarrow B = (A \Rightarrow B) \Rightarrow B = ((A \Rightarrow B) \Rightarrow B) \Rightarrow B' = \ldots$$

$$\cancel{\cfrac{\cfrac{\cfrac{A = (A \Rightarrow B)}{x : A \vdash x : A \Rightarrow B} \qquad \cfrac{}{x : A \vdash x : A}}{x : A \vdash x \cdot x : B}}{\mathrm{lam}\ x.\ x \cdot x : A \Rightarrow B}}$$

$$\cfrac{\cfrac{\cfrac{\cfrac{\cfrac{}{f : B \Rightarrow C, g : A \Rightarrow B, x : A \vdash f : B \Rightarrow C} \qquad \cfrac{\cfrac{}{f : B \Rightarrow C, g : A \Rightarrow B, x : A \vdash g : A \Rightarrow B} \qquad \cfrac{}{f : \_, g : \_, x : A \vdash x : A}}{f : B \Rightarrow C, g : A \Rightarrow B, x : A \vdash g \cdot x : B}}{f : B \Rightarrow C, g : A \Rightarrow B, x : A \vdash f \cdot (g \cdot x) : C}}{f : B \Rightarrow C, g : A \Rightarrow B \vdash \mathrm{la}\ x.\ f \cdot (g\ x) : A \Rightarrow C}}{f : B \Rightarrow C \vdash \mathrm{la}\ g.\ \mathrm{la}\ x\ f \cdot (g \cdot x) : (A \Rightarrow B) \Rightarrow A \Rightarrow C}}{\mathrm{lam}\ f.\ \mathrm{lam}\ g.\ \mathrm{lam}\ x.\ f \cdot (g \cdot x) : (B \Rightarrow C) \Rightarrow ((A \Rightarrow B) \Rightarrow A \Rightarrow C)}$$

$$\cfrac{\cfrac{\cfrac{}{f : Bool \Rightarrow Nat \vdash f : Bool \Rightarrow Nat} \qquad \cfrac{}{f : Bool \Rightarrow Nat \vdash \mathrm{nu}\ 1 + f \cdot \mathrm{true} : Bool}}{f : Bool \Rightarrow Nat \vdash f \cdot (\mathrm{nu}\ 1 + f \cdot \mathrm{true})\ Nat}}{\mathrm{lam}\ f.\ f \cdot (\mathrm{nu}\ 1 + f \cdot \mathrm{true}) : (Bool \Rightarrow Nat) \Rightarrow Nat}$$

$$(\mathrm{lam}\ x.x) \cdot \mathrm{nu}\ 1 = (x \mapsto x)[x \mapsto \mathrm{nu}\ 1] = \mathrm{nu}\ 1$$

$$\beta : (\mathrm{la}\ x.t) \cdot u = t[x \mapsto u]$$

$$\mathrm{comp} := \mathrm{lam}\ f.\ \mathrm{la}\ g.\ \mathrm{la}\ x.\ f \cdot (g \cdot x)$$

$$f, g, h \qquad (f \circ g) \circ h = f \circ (g \circ h)$$

$$\mathrm{comp} \cdot (\mathrm{comp} \cdot f \cdot g) \cdot h \overset{?}{=} \mathrm{comp} \cdot f \cdot (\mathrm{comp} \cdot g \cdot h)$$

$$= ((\mathrm{lam}\ x.\ \underline{\mathrm{la}\ y\ \mathrm{la}\ z.\ x \cdot (y \cdot z)}) \cdot (\mathrm{comp} \cdot f \cdot g)) \cdot h$$

$$=_\beta (\mathrm{lam}\ y.\ \mathrm{la}\ z.\ \mathrm{comp} \cdot f \cdot g \cdot (y \cdot z)) \cdot h$$

$$=_\beta \mathrm{la}\ z.\ \mathrm{comp} \cdot f \cdot g \cdot (h \cdot z)$$

$=$ ..... ②? $=$

$\boxed{\_5}$